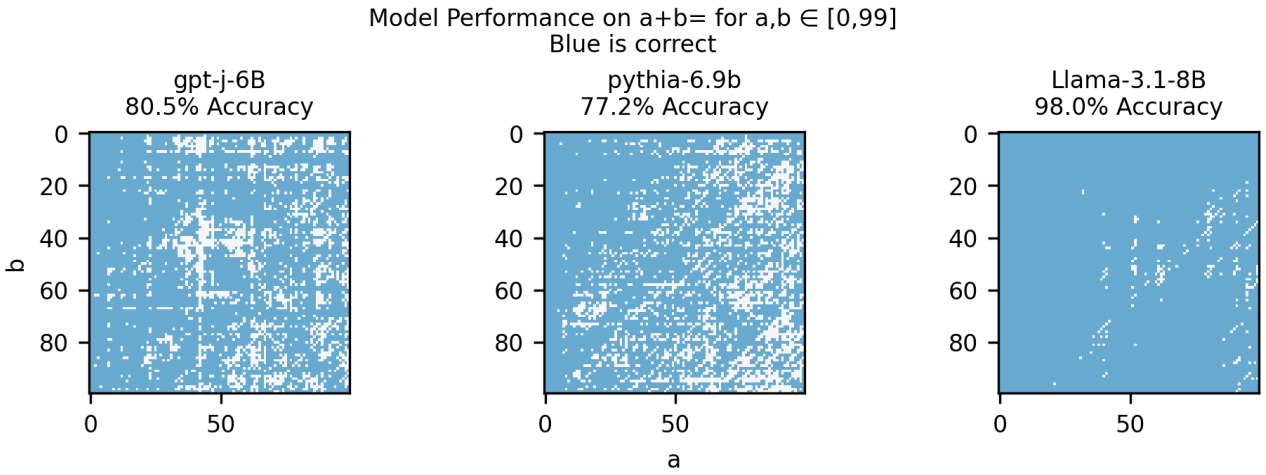


Figure 11. All models are able to competently perform the task $a + b$, with Llama3.1-8B performing best.

Table 2. The prompts used and accuracy of each model on the addition task $a + b$.

Model Name	**Prompt**	**Accuracy**
GPT-J	Output ONLY a number. $\{a\}+\{b\}=$	80.5%
Pythia-6.9B	Output ONLY a number. $\{a\}+\{b\}=$	77.2%
Llama3.1-8B	The following is a correct addition problem.\n$\{a\}+\{b\}=$	98.0%

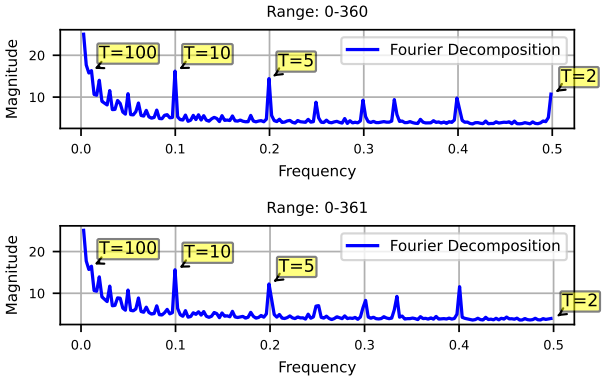


Figure 12. The $T = 2$ Fourier feature is prominent when analyzing h^0_{360}, but not when analyzing h^0_{361}.

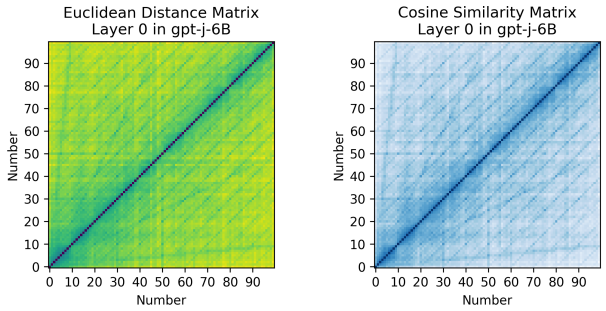


Figure 13. We see clear periodicity in GPT-J's layer 0 representations of the numbers from $0 - 99$.

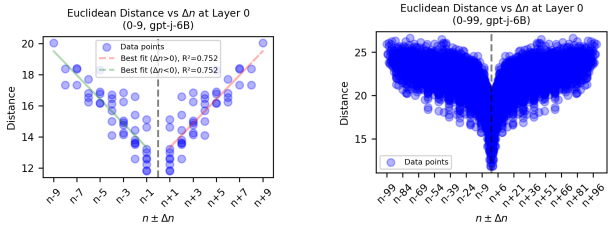


Figure 14. For GPT-J layer 0, the Euclidean distance between a and $a + \delta n$ is approximately linear for $a \in [0, 9]$, and sublinear for $a \in [0, 99]$. This provides additional evidence that GPT-J uses more than periodic features to represent numbers.

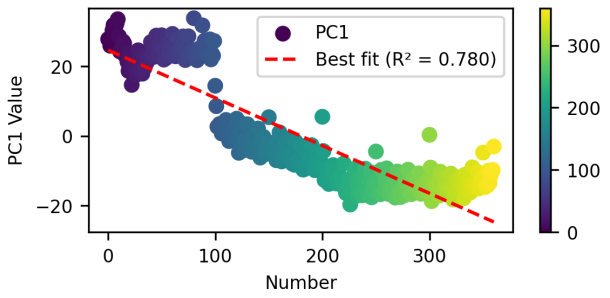


Figure 15. The first principal component of GPT-J layer 0 for numbers $a \in [0, 360]$ shows a discontinuity for three-digit a, implying that three-digit numbers are represented in a different space.

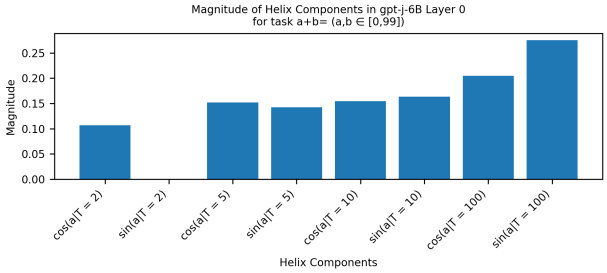


Figure 16. We plot the magnitude of each column of C, the coefficient matrix for the helical basis, to calculate the importance of each Fourier feature. Feature magnitude roughly increases with period, with the notable omission of the $T = 2$ sin component. We do not plot the linear component's magnitude because its output is a different scale.

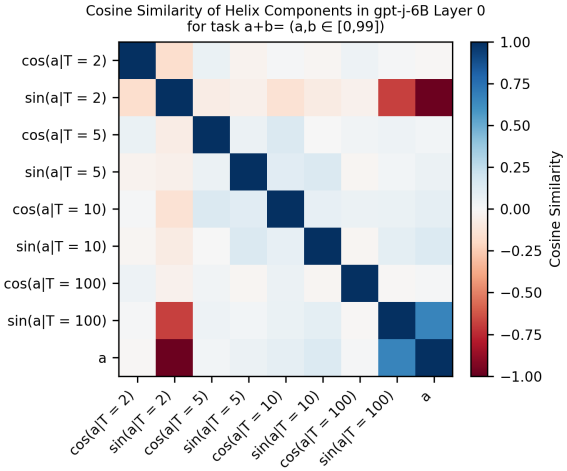


Figure 17. We plot the cosine similarity between columns of C, the coefficient matrix for the helical basis. Features are roughly orthogonal, which we expect for a helix, with the exception of the $T = 100$ sin component and the linear component a. We ignore the $T = 2$ sin component because of its negligible magnitude.

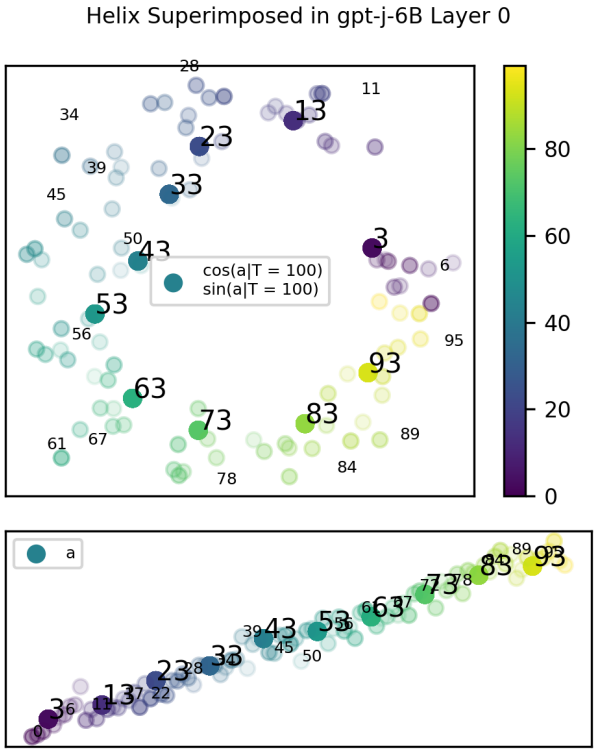


Figure 18. We fit all $a \in [0, 99]$ that do not end with 3 using a helix with $T = [100]$, and project the residual stream for $a = 3, 13, \ldots, 93$ onto the space. We find that there is continuity in the manifold, which we take as evidence that numbers are represented as a nonlinear feature manifold.

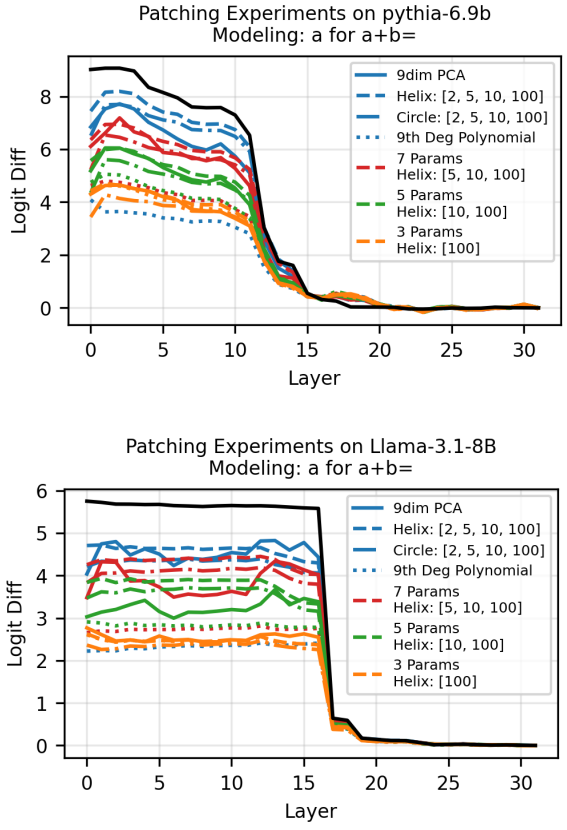


Figure 19. We replicate our patching results on Pythia-6.9B and Llama3.1-8B. The helical and circular fits outperform baselines at most numbers of parameters.

used and accuracy of GPT-J on these other tasks, see Table 3.

D. Additional Clock algorithm evidence

We show that $\text{helix}(a+b)$ fits last token hidden states for Pythia-6.9B and Llama3.1 8B in Fig. 23. Notably, the results for Llama3.1-8B are less significant than those for GPT-J and Pythia-6.9B. This is surprising, since the helical fit on the a token is causal for Llama3.1-8B in Fig. 19, and is a sign that Llama3.1-8B potentially uses additional algorithms to compute $a+b$. We hypothesize that this might be due to Llama3.1-8B using gated MLPs, which could lead to the emergence of algorithms not present in GPT-J and Pythia-6.9B, which use simple MLPs. Nikankin et al. (2024)'s analysis of Llama3-8B's top neurons in addition problems identifies neurons with activation patterns unlike those we identified in GPT-J. Due to this evidence, along with the importance of MLPs in the addition circuit, we consider it likely that Llama3.1-8B implements modified algorithms, but we do not investigate further.

D.1. Attention Heads

In Fig. 24, we use activation patching to show that a sparse set of attention heads are influential for addition. In Fig. 25, we find that patching in $k = 20$ heads at once is sufficient to restore more than 80% of the total effect of patching all $k = 448$ heads. In Fig. 26, we also find that all attention heads in GPT-J are well modeled using $\text{helix}(a, b, a+b)$. We judge this by the fraction of a head's total effect recoverable when patching in a helical fit.

We categorize heads as a, b, $a+b$, and mixed heads using a confidence score (detailed in Section 5.2). To make our categorization useful, we aim to categorize as few heads as mixed as possible. We find that using $m = 4$ mixed heads is sufficient to achieve almost 80% of the effect of patching the actual outputs of the $k = 20$ heads (Fig. 27), although using $m = 0$ mixed heads still achieves 70% of the effect. In Fig. 28, we analyze the properties of each head type. $a+b$ heads tend to attend to the last token and occur in layers 19 onwards. a, b heads primarily attend to the a and b tokens and occur prior to layer 18. Mixed heads attend to the a, b, and last tokens, and occur in layers 15-18.

To understand what each head type reads and writes to, we use a modification of the path patching technique we have discussed so far. Specifically, we view mixed and a, b heads as "sender" nodes, and view the total effect of each downstream component if only the direct path between the sender node and the component is patched in (not mediated by any other attention heads or MLPs). In Fig. 29, we find that both a, b and mixed heads generally impact downstream MLPs most. Similarly, we consider mixed and $a+b$ heads